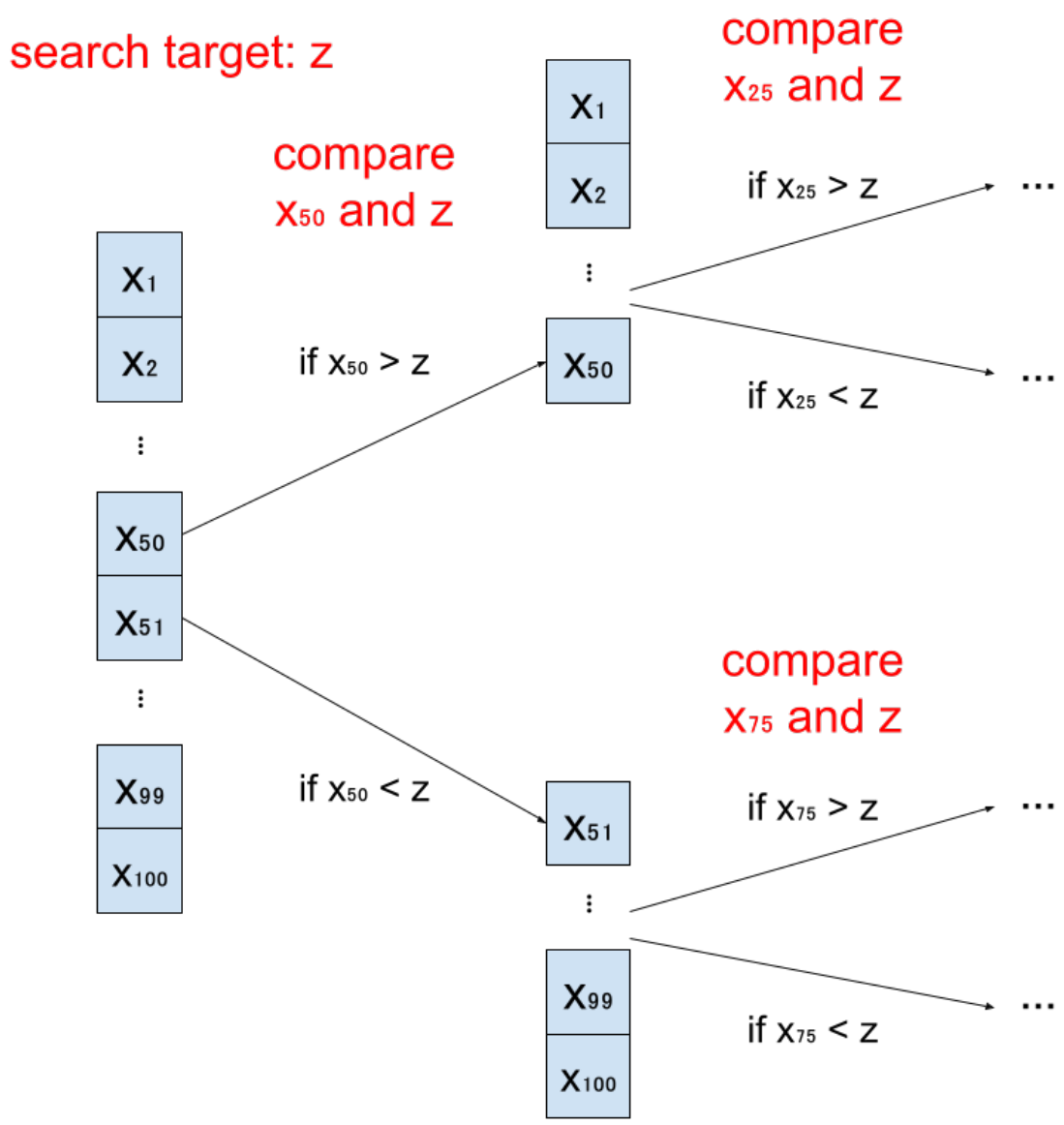


Figure 6.5 Illustration of binary search on a 100-element list. At each iteration of the search, the target is compared against the middle element of the current list. Depending on the result of this comparison, we remove either the first or second half of the list from the search space.

SIDEBAR **Terminating search early**

For each comparison in Figure 6.5, we omit the situation where z is equal to the number it's being compared to, in which we can simply terminate the search.

On average, this procedure helps us find the location of z within the list much faster than sequentially searching through the list from one end to the other.

NOTE In fact, the binary search strategy is optimal in this regard, allowing us to locate the location of z faster than any other strategy on average.

©Manning Publications Co. To comment go to liveBook
https://livebook.manning.com/#!/book/bayesian-optimization-in-action/discussion

Licensed to ashwin sampath <ashwin.manning@gmail.com>